Mobile App

Architecture & Scope of Work

Patient & Doctor · iOS & Android

Secure telemedicine platform for specialists and patients

Prepared for: 24Telemed · Document version 1.0

1. Executive Summary

> A single cross-platform mobile app (iOS & Android) that lets **patients sign up and consult doctors over secure video calls**, and lets **doctors receive and manage those consultations**. It reuses the existing 24Telemed Django backend — no rebuild — and adds the mobile experience on top.

The app ships **both roles in one binary**, selected at login/sign-up: **Patient** (self-service) and **Doctor**. Patients register themselves, complete their medical records, and either **call an available doctor instantly** or **book an appointment** for later. Doctors register, are **approved by an admin**, then receive incoming calls and scheduled consultations with live video.

2. Goals & Principles

- **Reuse the existing backend** — same database, authentication, video provider and call logic as the web apps.
- **Self-service** — patients and doctors onboard themselves; admins only approve doctors.
- **Native experience** — real device camera/microphone for high-quality video, not an embedded web page.
- **Secure by default** — encrypted token storage, doctor verification, role-based access.
- **No disruption to web** — all backend changes are additive and backward-compatible; the existing web apps are untouched.

3. User Roles

ROLE	WHO	ONBOARDING	CAN DO
Patient	Members of the public seeking care	Self sign-up, instant access	Manage medical records, call a doctor, book appointments, view history
Doctor	Licensed practitioners / specialists	Self sign-up → **admin approval required**	Receive & answer calls, run video consultations, manage bookings, edit profile
Admin	24Telemed staff	Existing Django admin (web)	Approve/verify doctors, manage users & data

Note: the previous “Health Personnel/Assistant” role is replaced by self-service Patients in the mobile app, per the updated requirement. The backend assistant role still exists and is unaffected.

4. Feature Scope

4.1 Onboarding & Account (both roles)

- **Sign up** as Patient or Doctor — name, contact, password, and (doctors) specialty.
- **Terms of Use & Privacy Policy** — must be accepted at sign-up; readable in-app.
- **Login** with a Patient/Doctor toggle; sessions persist securely between launches.
- **Change password** in-app; **forgot password** via email OTP (backend supported).
- **Animated branded splash** and consistent visual identity.

4.2 Patient Experience

- **Home dashboard** — greeting, live “doctors available now”, quick actions, upcoming appointment, available-doctor carousel, recent consultations, health tips.
- **Medical records** — create & edit a full personal health record: personal details, vitals (blood type, weight, height), allergies, current medications, chronic conditions, past & family medical history, immunisation record.
- **Call a doctor** — pick an online doctor, set priority & reason, and start an instant video consultation.
- **Book an appointment** — choose a doctor, day, time slot, priority and reason for a scheduled consultation.
- **My bookings** — view status (pending / confirmed / declined), cancel, and start when it’s time.
- **Consultation history** — past calls with the doctor, status and timing.
- **Account** — edit records, change password, view legal docs, sign out.

4.3 Doctor Experience

- **Dashboard** — availability status and call statistics (total call time, completed, busy, failed) plus recent consultations.
- **Incoming call** — full-screen ringing with patient context; **Answer** or **Decline**.
- **Video consultation** — native call with camera/mic controls and end-call.
- **Bookings** — review requests, **confirm** or **decline**, and **start** at the scheduled time.
- **Profile** — edit name, contact, specialty, location and bio; change password.
- **Approval gate** — until an admin verifies the account, the doctor sees an “awaiting approval” notice and is hidden from patients.

4.4 Live Video Consultation (shared)

- **Real-time signalling** — calls ring instantly using a persistent connection; presence shows who is online.
- **Native video** — built on the same VideoSDK video provider the web uses; both sides join the same secure room.
- **In-call controls** — mute/unmute, camera on/off, end call; automatic close when the other side hangs up.
- **Lifecycle** — call status flows from ringing → in-progress → completed and is recorded in history.

5. Technical Architecture

Mobile app

- Expo SDK 54 / React Native (iOS & Android, one codebase)
- TypeScript, file-based navigation (Expo Router)
- React Query for data fetching & caching
- Encrypted token storage (Secure Store)
- Native WebRTC video (VideoSDK)

Backend (reused)

- Django REST API + JWT authentication
- WebSockets (Django Channels + Redis) for call signalling
- VideoSDK rooms for the actual video
- Existing data: Users, Patients, Call Logs, Wallet
- Shared OpenAPI contract — same as the web apps

5.1 How a call works (end to end)

```
Patient                         Backend                          Doctor
   │  online (WebSocket)           │              online (WebSocket)  │
   │ ──── call doctor ───────────▶ │ ── ring incoming call ──────▶ │
   │                               │ ◀───────── answer ──────────── │
   │ ◀── room created + joined ─── │ ── room created + joined ───▶ │
   │  ══════════ live native video (VideoSDK room) ══════════      │
   │ ──── end ───────────────────▶ │ ── status: completed ───────▶ │
                     (recorded in consultation history)
```

5.2 Security & access control

- JWT access/refresh tokens stored in the device secure enclave; auto-restored on launch.
- Role-gated areas (patient vs doctor) enforced in-app and by the API.
- **Doctors are hidden and uncallable until an admin verifies them.**
- Camera & microphone used only during an active consultation, with OS permission prompts.

6. Backend Additions (for mobile — additive & web-safe)

ADDITION	WHY	IMPACT ON WEB
Public **self-registration** endpoint	Patients & doctors sign themselves up	None (web never calls it)
Doctor **verification flag** + admin approval	Gate doctors until approved	None (existing users default to verified)
Appointments / booking module	Schedule consultations for later	New, isolated module
Patient profile auto-linked to its account	Patients own their record	None (only affects self-service patients)

7. Build, Environments & Delivery

- **Development builds** for testing on real devices (required for native video).
- **Store releases** — Apple App Store & Google Play submission-ready builds via Expo EAS.
- **Environments** — switchable API targets (local / staging / production) via configuration.
- **Branding** — app icon, splash and theme aligned to the 24Telemed identity.

8. Feature Status & Options

Use this list to confirm scope — tick to keep, strike to remove, or request additions.

CAPABILITY	STATUS	NOTES
Patient self sign-up + login	Built	With Terms/Privacy acceptance
Doctor self sign-up + admin approval	Built	Verification gate in admin
Patient medical records (full)	Built	All backend health fields
Call a doctor (instant, native video)	Built	Real-time ring + VideoSDK
Book & manage appointments	Built	Confirm / decline / start
Doctor dashboard, incoming calls, profile	Built	Stats + ring/answer
Consultation history	Built	Both roles
Change password	Built	In-app
Terms of Use & Privacy Policy screens	Placeholder	Awaiting final legal text from client
Forgot-password (email OTP)	Backend ready	Mobile screen can be added
Wallet top-up & payments (Paystack)	Optional	Backend exists; needs public key + UI
Push notifications for incoming calls (app closed)	Optional	Recommended for reliability
Prescriptions / medical encounter notes	Optional	Backend supports; UI to be built
Profile & record photo upload	Optional	Image upload to storage
Doctor ratings & reviews	Optional	New feature
In-app chat / messaging	Optional	New feature
Appointment reminders	Optional	Push/email reminders
Genotype & extended record fields	Optional	Small backend field additions
Multi-language support	Optional	Localisation

9. Assumptions & Dependencies

- The existing Django backend, VideoSDK account, and Redis remain available and reachable.
- Final **Terms of Use** and **Privacy Policy** text to be provided by the client.
- Apple Developer & Google Play accounts required for store distribution.
- Native video requires a development/production build (not the generic Expo Go sandbox).
- Doctor licensing/credential checks are an operational (admin) process, supported by the verification gate.